security of the blockchain, and also provides proof of existence for transactions and events. You can access the information stored on the blockchain to support operations at the application level. To illustrate these concepts, we'll explore an improved version of the blind auction application that was introduced in chapter 5.

As you've seen, the *kinds* of on-chain data are strictly controlled by the blockchain protocol, with the *actual data values* being determined by the Txs originating from the applications. You must be aware of this limitation when designing your blockchain-based systems. For the blind auction use case, we'll focus on one of the on-chain data elements: the event logs stored in the receipt tree. Let's begin by considering the following:

- What an event is
- How to define an event
- How to emit an event
- How to use event logs in the Tx receipts to provide notifications to the user via the UI

6.2.1 On-chain event data

Events are notifications that can be emitted from functions to indicate the presence of a condition or flag during a smart contract function's execution. Solidity provides features to define and emit an event with and without parameters. Events are logged on-chain, in the receipt tree, and can be accessed by their names.

You can define an event anywhere in the smart contract code before you use it. Still, it's a good idea to set aside a standard location to define the events in your design and coding so that